Yeast Fermentation and Fuel Sources

Student worksheet | Biology 2e, Chapter 7 | 45-60 minutes

Before you begin: the gas-collection apparatus

A flask holds the mixture. A fitted stopper and tube connect it to a gas syringe. Gas entering the barrel can move its freely sliding plunger; the scale measures collected volume in milliliters. A water bath keeps samples at a consistent temperature.

In a real setup, check for leaks and never block the syringe plunger. Use fresh material for every independent trial. Before testing, predict how the available sugar might affect the measurements; record your reasoning on the next page.

Run a controlled comparison

For the physical lab, use the method-specific classroom worksheet. The revised target is 20 g yeast and 80 g sugar per 1,000 mL total water, with separate stocks and up to 60 minutes of balloon observations. Its stronger recipe differs from the virtual model below.

The virtual lab guides you through three trials with no added sugar, three with glucose, and three with sucrose. Each comparison uses 0.5 g yeast, 50 mL final mixture, and a 30 C water bath. Sugar treatments use 2 g sugar; the control has no added sugar.

1. In Collect observations, select the assigned treatment if it is not already highlighted. Click Prepare mixture. The flask stays empty until preparation is complete.

2. Click Connect gas syringe to fit the stopper and tubing. The gas syringe must be connected before the trial starts. Its plunger remains free to move.

3. Click Start trial. Observe for 30 simulated minutes. You can pause, change timer speed, advance five minutes, or run the remaining time. The app records gas volume every five simulated minutes; copy the readings into your observation table.

4. Review the readings, then click Collect data and continue. This saves the results and advances to the next trial with a fresh flask. Repeat until all nine trials are collected.

After the controlled comparison, choose Explore other settings. The exploration activity is at the end of the lab worksheet; your nine comparison trials remain saved separately.

Yeast Fermentation and Fuel Sources

Student worksheet / prediction | Biology 2e, Chapter 7 | 45-60 minutes

Name: _____ Class: _____ Date: _____

1. Predict and plan

How does the available sugar affect gas accumulation in a yeast culture?

Prepare fresh mixtures with no added sugar, glucose, and sucrose. Connect each flask to a gas syringe, observe for 30 simulated minutes, and collect three trials per treatment. Then explore temperature or a no-yeast control separately. Compare average gas accumulation rates over the same time interval.

Prediction and scientific reasoning:

Variables and controls, or sampling plan:

Keep consistent: Virtual model: 0.5 g baker's yeast, 50 mL final mixture, 30 deg C water bath, identical gas collection apparatus, 30-minute observation. Sugar treatments have equal mass concentration, not equal molarity.

Select your route

Part	Virtual / Classroom	Equipment or conditions
No added sugar		
Glucose		
Sucrose		

Use the same questions for either route. Always label the source of each dataset.

Yeast Fermentation and Fuel Sources

Student worksheet / observations | Biology 2e, Chapter 7 | 45-60 minutes

2. Collect observations

Record collected gas (ml). Repeat each treatment with fresh material for three independent trials.

Treatment / trial	0 min	5 min	10 min	15 min	20 min	25 min	30 min
No added sugar / 1							
No added sugar / 2							
No added sugar / 3							
Glucose / 1							
Glucose / 2							
Glucose / 3							
Sucrose / 1							
Sucrose / 2							
Sucrose / 3							

Source (virtual/classroom) for each treatment: _____

Trial summaries

Treatment	Trial 1	Trial 2	Trial 3	Mean
No added sugar				
Glucose				
Sucrose				

Metric: Average gas accumulation rate in mL/min = (volume at 30 min - volume at 0 min) / 30.

Observations and graph

Graph time on the x-axis and the recorded measurement on the y-axis on separate graph paper, or use the app. Label treatments, units, and data sources. Record unusual observations below.

Yeast Fermentation and Fuel Sources

Student worksheet / explanation | Biology 2e, Chapter 7 | 45-60 minutes

3. Explain the evidence

Average gas accumulation rate = (final volume - initial volume) / elapsed minutes. Gas from yeast is consistent with CO2 production, but volume alone does not prove fermentation rather than aerobic metabolism or directly measure ATP yield.

1. Compare the mean gas accumulation rates for the three treatments. Use numerical evidence and discuss overlap between trials.

2. Why include yeast with no added sugar? What additional control would help separate yeast activity from physical gas expansion?

3. Explain how fermentation supports continued glycolysis. Why can gas volume alone not establish the amount of ATP made or prove oxygen was absent?

4. Identify one limitation of equal-mass sugar comparisons or gas collection, and propose a better experiment.

Before sharing: check units, source labels, calculations, and evidence. The app saves on this device only; download or print your report.

Yeast Fermentation and Fuel Sources

Student worksheet / exploration | Biology 2e, Chapter 7 | 45-60 minutes

4. Explore after the controlled comparison

After collecting all nine trials, choose Explore other settings. Select a fuel and prepare a fresh mixture, then connect the gas syringe. Predict an outcome before changing a setting.

Explore bath settings from 15 to 40 C while the timer runs. The bath temperature changes gradually. After the timer pauses at 30 minutes, resume for up to 120 minutes or until the syringe reaches its 100 mL capacity. Use Fresh mixture / restart trial to test another fuel or turn off Include 0.5 g baker yeast to try a no-yeast control.

My question, prediction, and reasoning:

My change (fuel, yeast present or absent, temperature, and simulated time):

What happened immediately and later? Use measurements to compare your observations with your prediction.

Exploration does not change the comparison means. Download your data before restarting. This simplified model omits gas expansion, yeast damage, and substrate depletion; do not treat model times or temperature responses as measurements of real yeast.

Cut & sort: matter and energy

Optional introduction or extension | 10-15 minutes | Biology 2e, chapters 7 and 8.

Choose Card Activity on the lab overview to open the separate circular photosynthesis and respiration model, or arrange the printed cards on chart paper. Use two molecule sets for both halves at once, or reuse one 19-card set as you switch processes.

Print cutout pages in color, single-sided, at 100% scale. Cut along dashed lines. Use scissors safely. Group work: one person arranges, one counts atoms, and one explains; then rotate roles.

Model aerobic cellular respiration. Reuse the molecule cards from the photosynthesis sheets and add the three respiration symbols. Set aside any symbols not needed.

Count carbon, hydrogen, and oxygen atoms separately on each side. Explain how your arrangement shows conservation of matter and the transfer of energy. Energy symbols are annotations, not part of this simplified atom inventory.

This equation models aerobic respiration, not yeast fermentation. Fermentation does not use oxygen. Respiration begins in the cytoplasm; most later stages in plant and animal cells occur in mitochondria. Plants respire too.

Name: _____ Class: _____ Date: _____

My equation and symbol placement:

_____

Atom counts: Reactants C ____ H ____ O ____ | Products C ____ H ____ O ____

Explain where energy enters, is stored, or leaves this process:

_____

_____

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

O_2



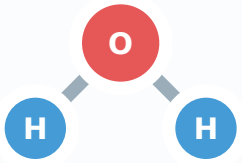
Oxygen

CO_2



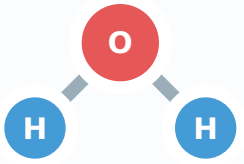
Carbon dioxide

H_2O



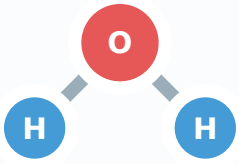
Water

H_2O



Water

H_2O



Water

O_2



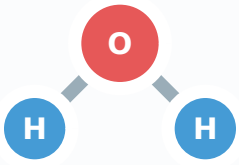
Oxygen

CO_2



Carbon dioxide

H_2O



Water

CO_2



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 1/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

+

Combine with

O_2



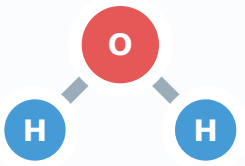
Oxygen

CO_2



Carbon dioxide

H_2O

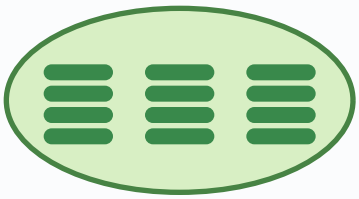


Water

+

Combine with

CHLOROPHYLL



Green pigment in chloroplasts
Simplified chloroplast diagram

Light-absorbing pigment

$C_6H_{12}O_6$



Atom inventory (not structure)

Glucose

O_2



Oxygen

LIGHT ENERGY



Sunlight

Atom colors: C = gray • O = red • H = blue

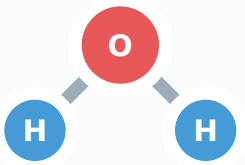
Student cutouts 2/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

H_2O



Water

→

Reaction arrow

O_2



Oxygen

O_2



Oxygen

CO_2



Carbon dioxide

CO_2



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

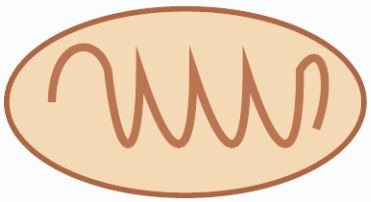
Student cutouts 3/3 • Print in color, single-sided, at 100%.

CELLULAR RESPIRATION

Add-on cutouts • Use with your photosynthesis molecule cards

Cut on dashed lines. Each picture is a symbol, not a measured quantity.

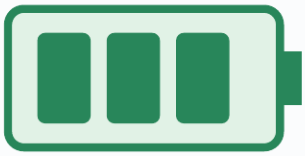
MITOCHONDRION



Most stages occur here

Simplified organelle diagram

ATP



Energy transferred to ATP

Usable energy for cell work

Battery = symbol for energy storage

HEAT



Energy released as heat

Transferred to the surroundings

Arrange these symbols with the molecule cards. Explain each placement.

Reuse the molecule cards; set aside symbols not needed for this process.